MIL-C-36066B(DMSC)
27 March 1962
SUPERSEDING
MIL-C-36066A(BuMed)
19 July 1961

MILITARY SPECIFICATION

CHLOROQUINE AND PRIMAQUINE PHOSPHATES TABLETS

1. SCOPE

1.1 This specification covers Chloroquine and Primaquine Phosphates Tablets suitable for oral use.

1.2 Classification. Chloroquine and Primaquine Phosphates Tablets shall be furnished in the following types of immediate containers, as specified (see 6.1):

Type I - Bottle (See 5.1.1.1).
Type II - Package (See 5.1.1.2).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566 Boxes, Folding, Paperboard.
PPP-B-585 Boxes; Wood, Wirebound.
PPP-B-601 Boxes, Wood, Cleated-Plywood.
PPP-B-621 Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 Box, Fiberboard.
PPP-B-676 Boxes, Set-Up, Paperboard.
PPP-C-186 Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

Military

MIL-L-10547 Liners, Case, Waterproof.

FSC 6505

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STANDARDS

Federal

Fed. Std. No. 00140 Tablets (For Medicinal Purposes).

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Food and Drug Administration - Federal Food, Drug, and Cosmetic Act and General Regulations for Its Enforcement.

(Application for copies should be addressed to the Food and Drug Administration, Washington 25, D. C.).

U.S. PHARMACOPOEIAL CONVENTION

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Printing Company, Easton, Pa.).

3. REQUIREMENTS

3.1 Material. Shall be Chloroquine and Primaquine Phosphates Tablets and shall be in accordance with all applicable requirements of Interim Federal Standard No. 00140 (Navy-BuMed), together with the additions and deletions stated herein:

S2 Classification. Shall be type I, class 2, style A, Grade 3. Grade 3 shall be plastic or ethyl cellulose film.

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S5.3 The following additional requirements are added to this paragraph:

Tablets shall contain 0.5 Gm (7-1/2 gr) of Chloroquine Phosphate (equivalent to 0.3 Gm (4-1/2 gr) of Chloroquine base) and 0.079 Gm (1-1/4 gr) of Primaquine Phosphate (equivalent to 0.045 Gm (11/16 gr) of Primaquine base), within the designated assay limits of the tablets.

Tablets shall assay to contain not less than 93.0 percent and not more than 107.0 percent of the required amount of Chloroquine Phosphate and not less than 93.0 percent and not more than 107.0 percent of the required amount of Primaquine Phosphate, when assayed by the methods specified in 4.3.1. Assay results shall be determined based upon the required metric quantity in the tablets.

Tablets shall comply with the identity tests specified in 4.3.2.

Chloroquine Phosphate and Primaquine Phosphate. The Chloroquine Phosphate and the Primaquine Phosphate used in the manufacture of the tablets shall be in accordance with the tests, standards, and requirements of the U.S.P.

All other ingredients used in the manufacture of the tablets shall comply with S5.1.

The taste of the drugs in the tablets shall be disguised by the film coating so that when one (1) tablet is kept in the mouth for 15 seconds, there shall be no taste of the drugs nor any other unpleasant taste.

S6.4 Color. Tablets shall be orange.

S6.4.9.1 The total disintegration time for the film-coated tablets shall be not more than 1 hour.

S6.14.1 Net contents. Net contents in type I, bottle, shall be 500 tablets, within the limits permitted in this paragraph. Contents in type II, package, shall be six (6) tablets. No tolerances shall be permitted.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government for a period of 3 years after delivery of supplies. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Sampling shall be conducted in accordance with Interim Federal Standard No. 00140.

4.3 Tests.

4.3.1 Assay. Tablets shall be assayed, as follows:

Determine the average weight of 20 Chloroquine and Primaquine Phosphates tablets and reduce to a fine powder. Accurately weigh about 0.75 Gm of this ground mixture into a 500-ml, glass-stoppered, volumetric flask; add some water and 5 ml of N/1 HCl; swirl to mix, then adjust to volume with water and mix well. Filter a portion through Whatman #40 paper and prepare the following dilutions:

Dilution #1 - 2 ml filtrate + 2 ml of N/1 HCl diluted to 200 ml with water.

Dilution #2 - 5 ml filtrate + 1 ml of N/1 HCl diluted to 100 ml with water.

Both dilutions shall be mixed well and then run on a Beckman spectrophotometer, or equivalent instrument, and compared against Standard Primaquine Phosphate solution and Standard Chloroquine Phosphate solution, using a N/100 HCl solution as the blank.

The Standard solutions shall be prepared to contain equivalent concentrations of Chloroquine Phosphate and of Primaquine Phosphate as that to be determined in the assay.

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The Chloroquine Phosphate and the Primaquine Phosphate shall be dissolved separately in the same solvent as that of the final sample dilutions.

Dilution #1 shall be read at 343 mu.
Dilution #2 shall be read at 281 mu.

The Chloroquine Phosphate Standard shall be read at 281 mu and at 343 mu.

The Primaquine Phosphate Standard shall be read at 281 mu and at 343 mu.

The weight, in mgs, of Chloroquine Phosphate in the original sample taken =

$$\frac{(K_1)\ (A_{343\ mu})\ (50{,}000) - (K_1')\ (A_{281\ mu})\ (10{,}000)}{K_1\ K_2' - K_1'\ K_2}$$

The weight, in mgs, of Primaquine Phosphate in the original sample taken =

$$\frac{(K_2')\ (A_{281\ mu})\ (10{,}000) - (K_2)\ (A343\ mu)\ (50{,}000)}{K_1\ K_2' - K_1'\ K_2}$$

where

K_1 = Absorbance per mg for Standard Primaquine Phosphate at 281 mu.

K_2 = Absorbance per mg for Standard Chloroquine Phosphate at 281 mu.

K_1' = Absorbance per mg for Standard Primaquine Phosphate at 343 mu.

K_2' = Absorbance per mg for Standard Chloroquine Phosphate at 343 mu.

A = Absorbance reading on sample dilutions (#1 and #2) at wavelengths specified.

Calculate the percent of required amount of the Chloroquine Phosphate and of the Primaquine Phosphate in the tablets.

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4.3.2 Identity. The solutions (dilutions) used in the assay procedure, specified in 4.3.1, shall give the characteristic absorption plateau at about 281 mu and a peak at 343 mu.

A filtered, distilled water solution of the tablets, 1 in 100, neutralized with ammonia T. S. (U.S.P.) shall give a yellow precipitate when silver nitrate, T.S., (U.S.P.) is added.

Digest a quantity of finely powdered tablet material, containing the equivalent of about 25 mg of Primaquine Phosphate, with 10 ml of water for 15 minutes, and filter. Dilute 0.1 ml of the filtrate with 1 ml of water, and add 1 drop of gold chloride T.S. A violet-blue color shall be produced at once.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery of the product shall be as specified herein and shall be in accordance with the applicable paragraphs of Federal Specification PPP-C-186, dated 11 December 1961.

5.1.1 Immediate containers.

5.1.1.1 Type I, bottle. Each type I bottle (containing 500 tablets) shall comply with the following classification:

GROUP A CLASS 1 TYPE e STYLE 2 GRADE 1

CLOSURE A or B SEAL A or B

5.1.1.2 Type II, package. Type II, package shall be in accordance with the requirements specified in 5.3.3.1.2.

5.2 Labeling. Labeling shall be as specified hereinafter. (See 5.2.1 for bottle of 500 tablets or 5.2.2 for package of six (6) tablets).

5.2.1 Type I, bottle.

5.2.1.1 Immediate containers. Each immediate container label shall bear the following information only:

(a) 500 Tablets

Chloroquine-Primaquine Tablets

Each tablet contains chloroquine phosphate, U.S.P., 500 mgm (7-1/2 gr), (equivalent to 300 mgm (4-1/2 gr) base and primaquine phosphate, U.S.P., 79 mgm (1-1/4 gr), equivalent to 45 mgm (11/16 gr) base).

(See additional labeling requirements on page 7)

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FOR MALARIA PROPHYLAXIS ONLY

Not to be used for the treatment of clinical malaria

Suggested adult dose: 1 tablet per week

(b) the stock number designated as "FSN 6505-753-5043" or "Stock No. 6505-753-5043"

(c) the lot or control number designated as "Lot No._____" or "Control_____"

(d) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

The label format shall be as stated above and those parts that are underlined shall be underlined.

5.2.2 Type II, package. Each type II package (containing six (6) tablets)) shall be labeled in accordance with the requirements of the Federal Food, Drug, and Cosmetic Act, and shall include the information required below:

On the front face of the flap of the unit package ("match folder" type carton):

(a) the item identification designated as "CHLOROQUINE AND PRIMAQUINE PHOSPHATES TABLETS"

(b) the phrase "6 tablets" or a similar phrase

(c) the stock number designated as "FSN 6505-854-2239" or "Stock No. 6505-854-2239"

(d) the lot or control number designated as "Lot No._____" or "Control_____"

(e) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

(f) the statement "Each tablet contains chloroquine phosphate, U.S.P., 500 mgm (7-1/2 gr), (equivalent to 300 mgm (4-1/2 gr) base and primaquine phosphate, U.S.P., 79 mgm (1-1/4 gr), equivalent to 45 mgm (11/16 gr) base).

(See additional labeling requirements on Page 8)

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(g) the statement "FOR PREVENTION OF MALARIA" or a similar statement.

(h) the statement "See reverse side for instructions" or a similar statement.

The statement "Caution: Federal law prohibits dispensing without prescription" shall not appear.

On the inner face of the flap of the unit package ("match folder" type carton):

INSTRUCTIONS FOR USE:
Take ONE TABLET PER WEEK during or immediately after meals. DO NOT CHEW. To be used in those areas where infection with malaria is possible. Following departure from such an area, take ONE TABLET PER WEEK for six weeks.

5.3 Packaging.

5.3.1 Unit of issue. One (1) bottle of 500 tablets, or one (1) package of 6 tablets, as specified, constitutes one unit of issue.

5.3.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each intermediate and exterior container, as applicable for the required level of protection specified in the procurement document.

Table I. Packaging quantities

Type	Unit package	Intermediate package	Exterior container
I	Not required	12 units	48 units
II	1 unit	25 units	600 units

5.3.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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5.3.3 Level A.

5.3.3.1 Unit package.

5.3.3.1.1 Type I, bottle. No individual cartoning shall be required.

5.3.3.1.2 Type II, package. Each unit (6 tablets) shall be sealed between two sheets of plastic coated foil or other suitable material in such a manner that each tablet shall be contained in an individual, watertight pocket. The sealed unit shall be securely attached to, and protected by, a "match folder" type carton.

5.3.3.2 Intermediate package.

5.3.3.2.1 Type I, bottle. Type I bottles shall be packaged in a carton of suitable size and design, constructed in accordance with Federal Specification PPP-B-636, type I, class 1. Carton design shall include partitions, liners, and top and bottom pads. Partitions shall be full-height, half-slotted style, fabricated of the same material as the carton. The partitions shall form an individual, snug-fitting cell for each immediate container. Liners shall be of one-piece, covering the sides and ends of the carton. Liners shall be the same height as the partitions and fabricated of the same material as the carton. Top and bottom pads fabricated of the same material as the carton, and not more than 1/16 inch smaller than the inside length and width of the carton, shall be positioned on bottom and top of liners and partitions. Closure shall be adequate to prevent spillage of contents under normal handling.

5.3.3.2.2 Type II, package. Type II packages shall be packaged in a carton of suitable size and design, constructed in accordance with Federal Specification PPP-B-566, PPP-B-636, or PPP-B-676. Commercial colors shall be acceptable for paperboard cartons. Closure shall be adequate to prevent spillage of contents under normal handling.

5.3.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

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5.4 Packing.

5.4.1 Level B.

5.4.1.1 Exterior container. Exterior container shall be a double-faced or double-wall corrugated fiberboard carton of suitable size and design, having a minimum bursting strength test of 275-pounds in accordance with Federal Specification PPP-B-636, type I, class 1. Closure shall be effected in accordance with the requirements of Federal Specification PPP-B-636.

5.4.2 Level A.

5.4.2.1 Exterior container. Subject item, packed to the degree of protection required for level B packing, shall be further protected by being overpacked in an exterior container designed for type 2 load and constructed in accordance with Federal Specifications PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; or PPP-B-621, table III.

5.4.2.2 Case liner. Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.4.2.3 Strapping. Strapping, when required, shall be in accordance with applicable box specification.

NOTE: Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistibution.

5.4.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in contract or purchase order.

5.5 Marking.

5.5.1 Intermediate package. Each intermediate package shall be marked in accordance with 5.2.of Military Standard MIL-STD-129. Waterproofing on labels shall be required only when the carton utilized is fabricated of water-resistant material. Lot (control) number, contract number, and name of contractor shall be shown.

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5.5.2 Exterior container. Exterior container shall be marked in accordance with Military Standard MIL-STD-129. Lot (control) numbers shall be shown. The letter "L" shall precede the Federal stock number. One space shall be allowed between the letter "L" and the first digit of the stock number. The terms "Stock number" or "FSN" shall not be indicated. Service color and service symbol markings are required only on direct shipments from contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Unit of issue, i.e., bottle or package.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

6.3 Transportation description. Transportation description applicable to this item is:

Drugs or medicines NOIBN

Carload minimum weight 24,000 pounds subject to Rule 34 of Uniform Freight Classification.

Truckload minimum weight 24,000 pounds per National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

Custodians:
Army-Med
Navy-BuMed
Air Force-AFCSG

Preparing activity:
Navy-BuMed
(DSMC)

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